

W5+H Architectural Matrix & Node Topology

Comprehensive Synthesis: Who, What, When, Where, Why, How

Derived from: W5/H-(6-files) (User explains audit.txt, Where and When.txt, What and Why.txt, What and How.txt, Copy of Agent explanation.txt) and 4-ARCHITECTURE-cont.-(6-files).

Status: Canonical Living Wiki Reference | Clean Markdown Layer

Target Repository: data_vault/02_wiki_md/

1. The W5+H Ecosystem Matrix

Dimension	Architectural Specification	Real-World Hardware & Subsystem
WHO	• Operator: Derek LeGrand (System Architect & Sole Deletion Authority). • Agent Personas: Horizons Concierge (Voice/UI), Æsc Terminal Daemon (Shell/ADB), Æyre Media Daemon (Sensory VAD/TTS), 0.8B Triage Router, 9B Executor, Home Assistant Auditor, and Incognito Red Gatekeeper.	Node Alpha (Motorola Razr Ultra 2025), Node Beta (Jetson Orin Nano Super), Node Gamma (Rubik Pi 3), Node Delta (GCP Cloud).
WHAT	• Sovereign, multi-tier cognitive agent ecosystem running local open weights and decoupled daemons without root or Termux sandbox limits. • #d.u.m.b.a.s.s. universal memory backbone across SQLite, PostgreSQL, mem0,	3-APK Android Suite, Local Unix Abstract Sockets, Tailscale Mesh, GCP Vertex AI.

Dimension	Architectural Specification	Real-World Hardware & Subsystem
	OB1, OmniRoute, and Reasoning Bank. • Continuous learning loop via asynchronous audit logs and RLVR training.	
WHEN	• Phase 1: Knowledge Curation & Universal JSONL Marker Indexing (Current). • Phase 2: Tool Harness Ingestion & Action Schemas. • Phase 3: Repository Layout & Multi-Agent Grill Sessions. • Phase 4: 3-APK Native Android Deployment. • Phase 5: Local Dual-Model Inference & Continuous Improvement RLVR Loop. • Phase 6: Multi-Node Mesh Orchestration & Autonomous Agent Swarm.	Phased roadmap spanning real-time mobile execution to scheduled cloud batch training.
WHERE	• Edge Controller: Node Alpha (Moto Razr Ultra 2025, Snapdragon 8 Elite / Hexagon v79 NPU, 16GB RAM). • Compute Core: Node Beta (NVIDIA Jetson Orin Nano Super 8GB, CUDA 60–70 TOPs, NVMe SSD). • Workstation: Node Gamma (Rubik Pi 3 Dragonwing, Dual-Monitor Ribbon Display Server). • Cloud Pipeline: Node Delta (Google Cloud Platform,	Peer-to-peer encrypted Tailscale mesh with local air-gapped Wi-Fi router switch fallback.

Dimension	Architectural Specification	Real-World Hardware & Subsystem
	Vertex AI \$1,000 credit tier, GCS buckets).	
WHY	• Eliminates mobile RAM saturation and Android Low Memory Killer (LMK) crashes by decoupling UI from execution daemons. • Eliminates catastrophic forgetting and prompt drift through reset-free Continual Harness adaptation and immutable base anchors. • Protects intellectual property by separating personal developer credits from business workspace storage buckets via asymmetric IAM.	Sovereign autonomy, zero data leakage, high-throughput local execution.
HOW	• Communication: Local UNIX domain abstract sockets (<code>\0</code>), Ktor/Netty WebSockets, and ADB local TCP loopback (<code>127.0.0.1:5555</code>). • Inference Routing: OmniRoute proxy on port 20128 applying RTK/Caveman token compression (15%–95% savings) with async memory extraction. • Memory: mem0 (in-session habit caching) + OB1 (static MCP retrieval) + Reasoning Bank (<code>active_execution_paths.json</code>).	Hardware acceleration (Hexagon HTP FastRPC + Jetson CUDA) and automated scripts.

2. Multi-Node Sovereign Mesh Topology

| NODE ALPHA: Motorola Razr Ultra 2025 (Snapdragon 8 Elite / Hexagon v79) |

- 16 GB RAM (8.0–11.5 GB carved out for local GGUF weights)

- Decoupled 3-APK Native Android Suite:

| — Horizons UI: Chromium WebView frontend & persistent WebSockets

| — Esc: System Terminal Daemon (Local ADB loopback 127.0.0.1:5555) |

└─ Æyre: Media Daemon (Silero VAD + Moonshine STT + Kokoro TTS)

- Models: Qwen 3.5 0.8B (Triage) + Qwen 3.5 9B (Executor)

- Memory: mem0 in-session episodic state

| (Encrypted Tailscale Mesh)



| NODE BETA: NVIDIA Jetson Orin Nano Super 8GB (Headless Ubuntu Server) |

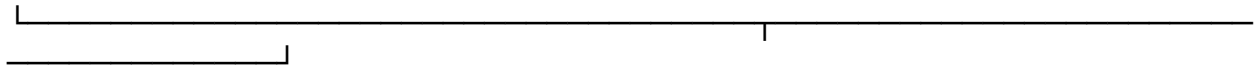
- Dedicated CUDA Core Array (60–70 TOPs), 500+ GB NVMe Storage

- #d.u.m.b.a.s.s. Core Hub: PostgreSQL 16+ with pgvector OB1 database |

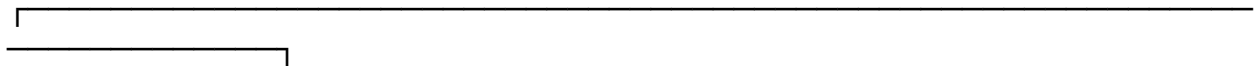
- Multi-Agent Home Swarm:

| — Home Assistant: Cross-Agent Auditor & daily log compiler

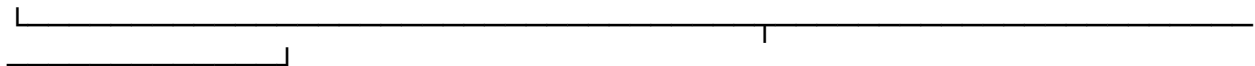
- | |— Red Agent Auditor: Air-gapped compliance & hallucination checker |
- | |— IT Help Desk Agent: Hardware specs & diagnostic runbooks |
- | |— NPU / Thermal Watchdog: Cluster headroom & task-shedding daemon |



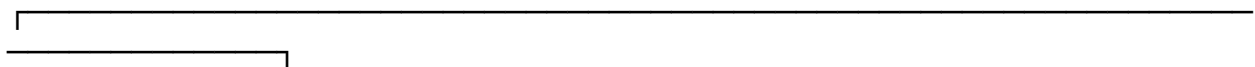
| (Encrypted Tailscale Mesh)



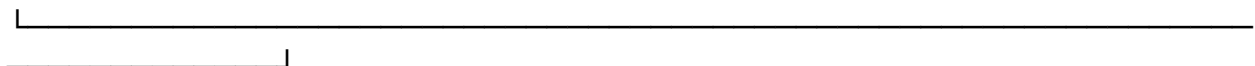
- | NODE GAMMA: Rubik Pi 3 Dragonwing (Qualcomm SoC / 14+ TOPs) |
- | - Dual-Monitor Ribbon Display Server & hardware keyboard hub |
- | - System log visualizer, telemetry dashboard & DroidDesk workstation |



| (Cloud Cross-Account IAM Handshake)



- | NODE DELTA: Google Cloud Platform (Enterprise Asymmetric Cloud) |
- | - \$1,000 Vertex AI Agent Builder credit pool (Personal Account) |
- | - gs://business-vault-bucket (Business Secure Workspace) |
- | - Heavy batch training, RLVR fine-tuning (Unsloth/GRPO), JSONL stream |



3. The Four-Part Auditing & Trapdoor Protocol

As detailed in [User explains audit.txt](#), the audit loop operates with four distinct components:

1. **On-Device Dual Agent:** The live interaction tandem (0.8B Triage + 9B Executor) generating step-by-step task executions.
2. **Main Computer Twin Agent (Node Beta):** Mirrors execution state, logs prompts, tool calls, and outputs into daily JSONL staging buffers.
3. **The Systems Secretary / Files Manager:** Packages candidate scripts, strips conversational boilerplate, and attempts to forward the batch to the cloud editor.
4. **The Incognito Red Gatekeeper (The Trapdoor):** Sits directly at the hand-off point. The files manager drops the batch into the queue, but the Red Agent intercepts it inside a sealed sandbox ([.incognito_red_sandbox/](#)), evaluating it against [nope_data_bank.json](#). If verified (+1.0), it moves to [approved/](#) for cloud RLVR training; if flawed (-1.0), it is quarantined with full error traces.